

BILKENT UNIVERSITY
 Department of Electrical and Electronics Engineering
 EEE102 Introduction to Digital Circuit Design
 MidTerm Exam I Solution
 14-03-2005
 Duration 110 minutes

1. (16 points)

- a) Show that $A \cdot B = A \cdot C$ does not imply that $B = C$.
- b) Show that NAND and NOR operators are not associative.
- c) Given that $X \cdot Y' + X' \cdot Y = Z$, show that $X \cdot Z' + X' \cdot Z = Y$.
- d) Convert the expression $\{[(A+B+A' \cdot C') \cdot C+D]' + A \cdot B'\}$ into sum of minterms.

Solution:

1a)

	A	B	C	$A \cdot B$	$A \cdot C$
0	0	0	0	0	0
1	0	0	1	0	0
2	0	1	0	0	0
3	0	1	1	0	0
4	1	0	0	0	0
5	1	0	1	0	1
6	1	1	0	1	0
7	1	1	1	1	1

Note that in rows 0,1,2,3,4, and 7, $A \cdot B$ and $A \cdot C$ have the same values, but in rows 1 and 2, B and C are not equal.

1b) If nand operator is associative then $[(A \text{ nand } B) \text{ nand } C]$ must be equal to $[A \text{ nand } (C \text{ nand } B)]$.

$$(A \text{ nand } B) \text{ nand } C = [(A \cdot B)' \cdot C]' = A \cdot B + C' \quad (1)$$

$$A \text{ nand } (C \text{ nand } B) = [A \cdot (B \cdot C)']' = A' + B \cdot C \quad (2)$$

Expressions (1) and (2) are not equal because forexample if $A = 0$ and $C = 1$, then expression (1) is 0 but expression (2) is 1.

If nor operator is associative then $[(A \text{ nor } B) \text{ nor } C]$ must be equal to $[A \text{ nor } (C \text{ nor } B)]$.

$$(A \text{ nor } B) \text{ nor } C = [(A+B)' + C]' = (A+B) \cdot C' \quad (1)$$

$$A \text{ nor } (C \text{ nor } B) = [A + (B+C)']' = A' \cdot (B+C) \quad (2)$$

Expressions (1) and (2) are not equal because forexample if $A = 0$ and $C = 1$, then expression (1) is 0 but expression (2) is 1.

1c) $XY' + X'Y = Z$ is given.

$$\begin{aligned}
 XZ' + X'Z &= X(XY' + X'Y)' + X'(XY' + X'Y) && \text{substituting } Z \\
 &= X(X' + Y)(X + Y') + X'XY' + X'X'Y \\
 &= X(X'X + X'Y' + YX + YY') + X'Y \\
 &= XY + X'Y \\
 &= (X + X')Y \\
 &= Y
 \end{aligned}$$

$$XY' + X'Y = Z \text{ means } Z = X \oplus Y$$

$$\begin{aligned}
 \text{Shorter solution: } XZ' + X'Z &= X \oplus Z = X \oplus (X \oplus Y) = (X \oplus X) \oplus Y \\
 &= 0 \oplus Y = Y
 \end{aligned}$$

$$\begin{aligned}
 \text{1d) } &[(A+B+A' \cdot C') \cdot C+D]' + A \cdot B' \\
 &= [AC+BC+D]' + AB' \\
 &= (A'+C')(B'+C')D' + AB' \\
 &= (A'B'+A'C'+B'C'+C')D' + AB' \\
 &= (A'B'+C')D' + AB' \\
 &= A'B'D' + C'D' + AB' \\
 &= A'B'(C'+C)D' + (A'B'+A'B+AB'+AB)C'D' + AB'(C'D'+C'D+CD'+CD) \\
 &= A'B'C'D' + A'B'CD' + A'B'C'D' + A'BC'D' + AB'C'D' + ABC'D' \\
 &\quad + AB'C'D' + AB'C'D + AB'CD' + AB'CD \\
 &= A'B'C'D' + A'B'CD' + A'BC'D' + AB'C'D' + ABC'D' \\
 &\quad + AB'C'D + AB'CD' + AB'CD
 \end{aligned}$$

Another method of solution: $F = [(A+B+A' \cdot C') \cdot C+D]' + A \cdot B'$ has the TT

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

We can then write the sum of minterms

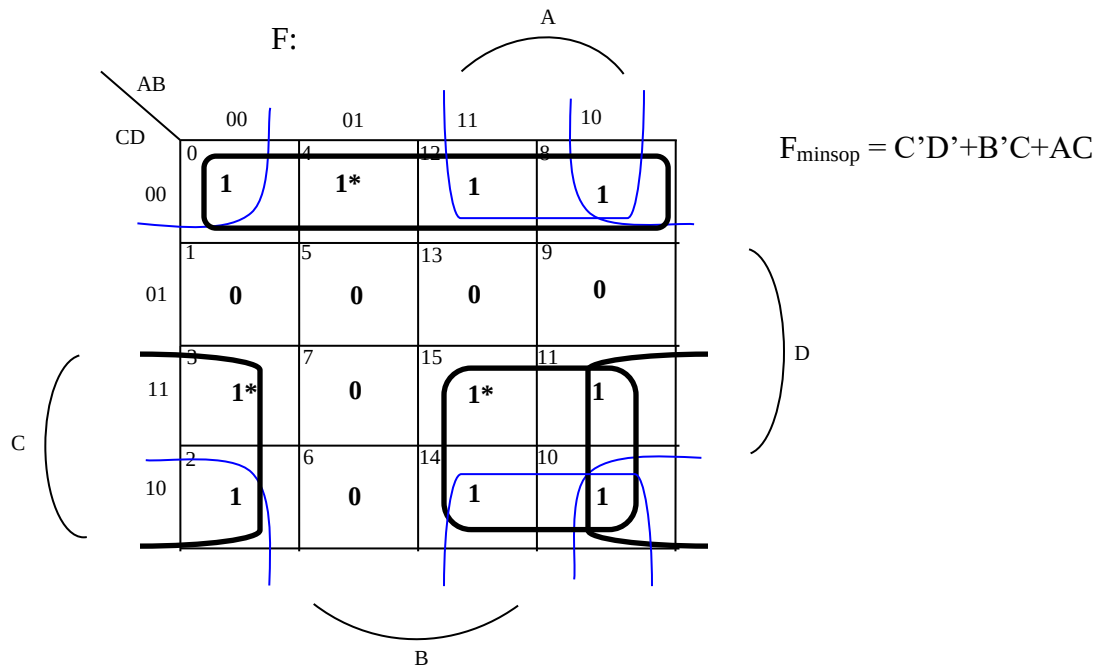
$$F = A'B'C'D' + A'B'CD' + A'BC'D' + AB'C'D' + AB'C'D + AB'CD' + AB'CD + ABC'D'$$

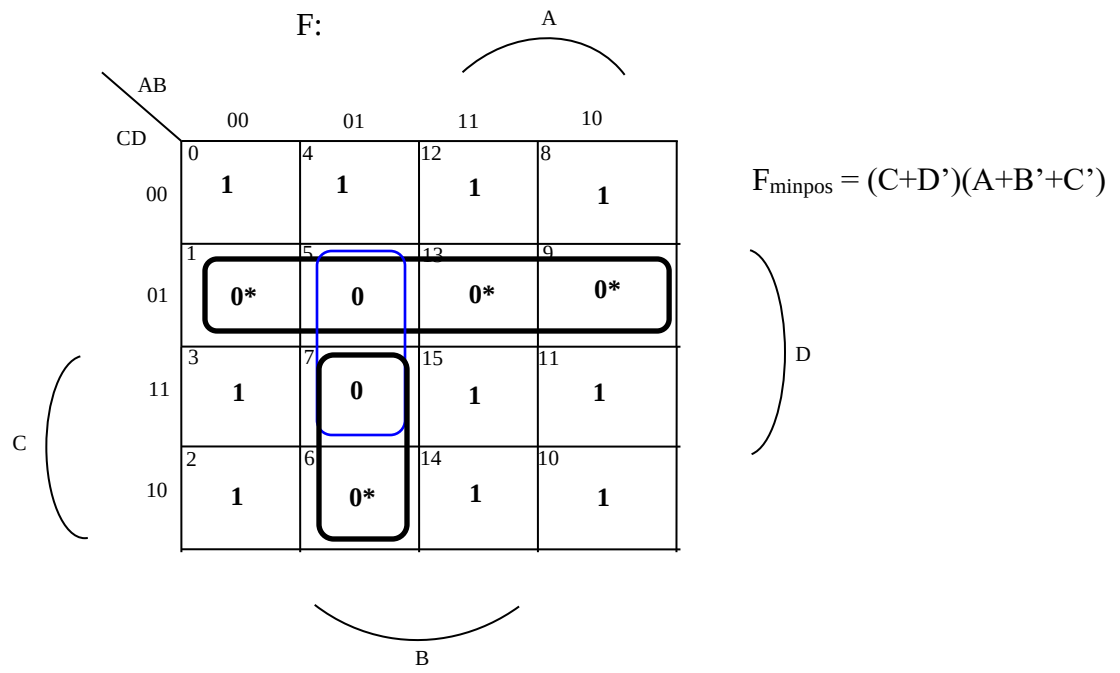
2. (12 points) Find all minimal sum expressions and all minimal product expressions for

$$F = \sum_{A,B,C,D} (0, 2, 3, 4, 8, 10, 11, 12, 14, 15).$$

(Draw Karnaugh Maps, indicate prime implicants and distinguished 1s and 0s).

Solution:





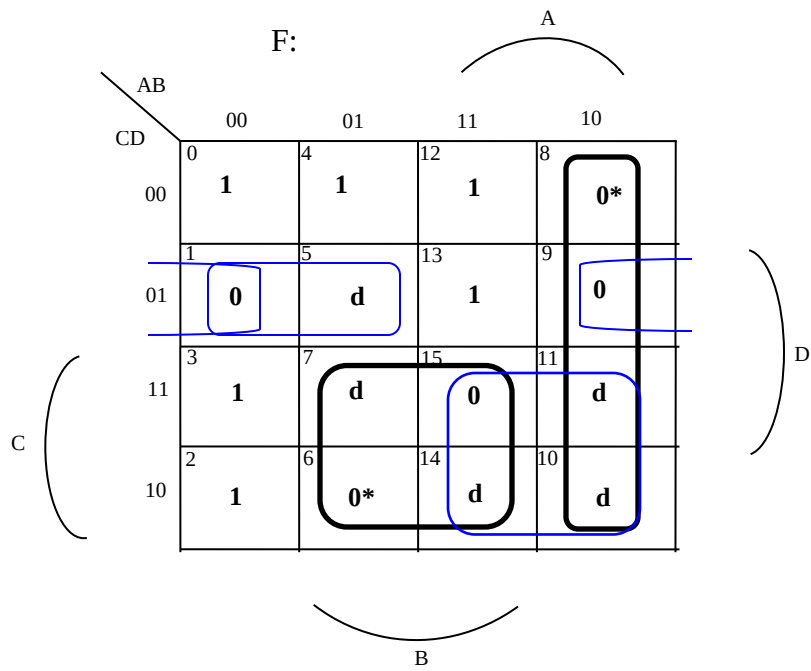
- $$F = \sum_{A,B,C,D} (0, 2, 3, 4, 12, 13) + d(5, 7, 10, 11, 14).$$

F:

	00	01	11	10
00	1	1	1	0
01	0	d	1*	0
11	1	d	0	d
10	1	0	d	d

5	7	10	11	14
1	0	1	1	0

5	7	10	11	14
1	0	1	1	0



$$F_{\text{minpos1}} = (B' + C')(A' + B)(A + C + D')$$

5	7	10	11	14
0	0	0	0	0

$$F_{\text{minpos2}} = (B' + C')(A' + B)(B + C + D')$$

5	7	10	11	14
1	0	0	0	0

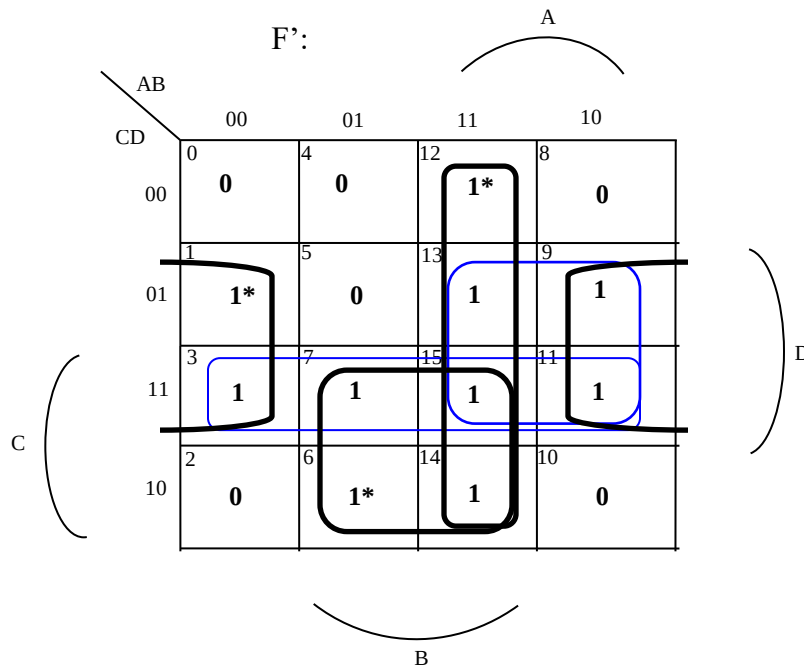
F_{minpos1} and F_{minpos2} are equivalent because don't cares have the same values. All the other solutions have different don't care values.

4. (12 points) Find all minimal sum expressions for

$$F = \prod_{A,B,C,D} (1, 3, 6, 7, 9, 11, 12, 13, 14, 15).$$

Also find all minimal product expressions for F by using the methodology of finding minimal sums.

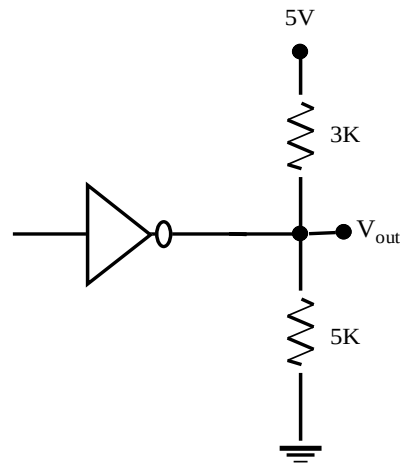
(Draw Karnaugh Maps, indicate prime implicants and distinguished 1s and 0s).



$$F'_{\text{minsop}} = B'D + AB + BC$$

$$F'_{\text{minpos}} = (B+D')(A'+B')(B'+C')$$

5. (16 points) A CMOS inverter working from 5V supply has a resistive load as shown below.



Suppose this CMOS inverter has the following specifications:

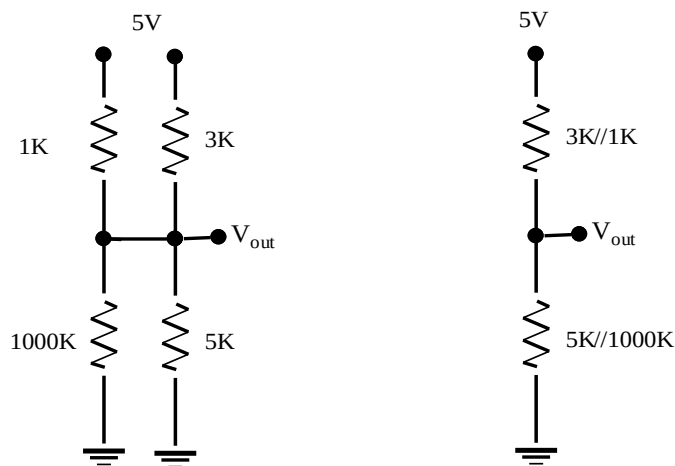
$$V_{IHmin} = 3.5 \text{ V}, \quad V_{ILmax} = 1.5 \text{ V}, \quad I_{ILmax} = -10 \mu\text{A}, \quad I_{IHmax} = 15 \mu\text{A}.$$

The ON and OFF resistances of the NMOS and PMOS transistors are $1\text{K}\Omega$ and $1000\text{K}\Omega$ respectively.

- Find V_{out} for when it is HIGH and also for when it is LOW.
- How many additional CMOS inverters of the same type can be connected to V_{out} ?

Solution:

a) Output HIGH:

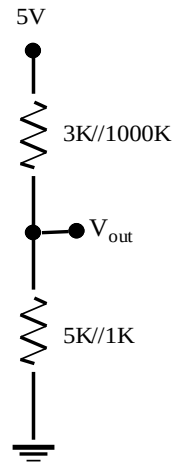
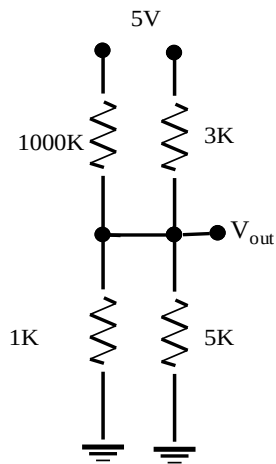


$$3K // 1K = 0.75K$$

$$1000K // 5K \approx 5K$$

$$V_{out} = 5V \times \frac{5}{5 + .75} = 4.35V$$

Output LOW:

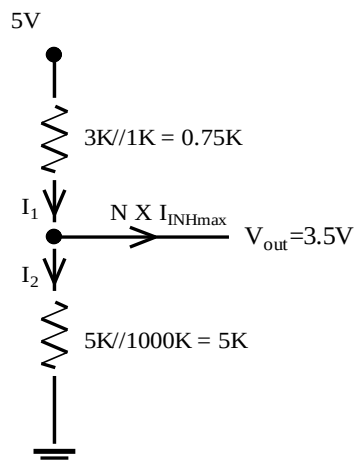


$$3K // 1000K \approx 3K$$

$$1K // 5K \approx 0.833K$$

$$V_{out} = 5V \times \frac{0.833}{3 + .833} = 1.09V$$

b) Output HIGH



$$V_{out} = 3.5V$$

$$I_1 = \frac{5-3.5}{0.75K} = 2mA$$

$$I_2 = \frac{3.5}{5} = 0.7mA$$

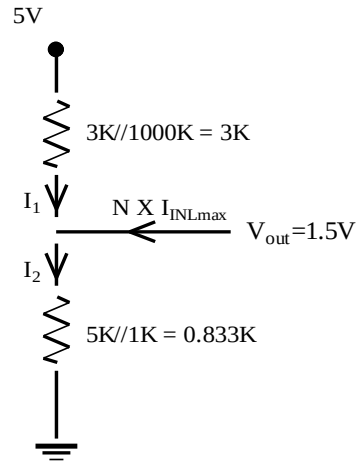
$$N \times 15\mu A = I_1 - I_2 = 2 - 0.7 = 1.3mA$$

$$N = \frac{1.3mA}{15\mu A} = 86.7$$

N must be an integer less than or equal to 86.7

Therefore take $N = 86$

Output LOW



$$V_{out} = 1.5V$$

$$I_1 = \frac{5-1.5}{3K} = 1.167mA$$

$$I_2 = \frac{1.5}{0.833} = 1.8mA$$

$$N \times 10\mu A = I_2 - I_1 = 1.8 - 1.167 = 0.633mA$$

$$N = \frac{0.633mA}{10\mu A} = 63.3$$

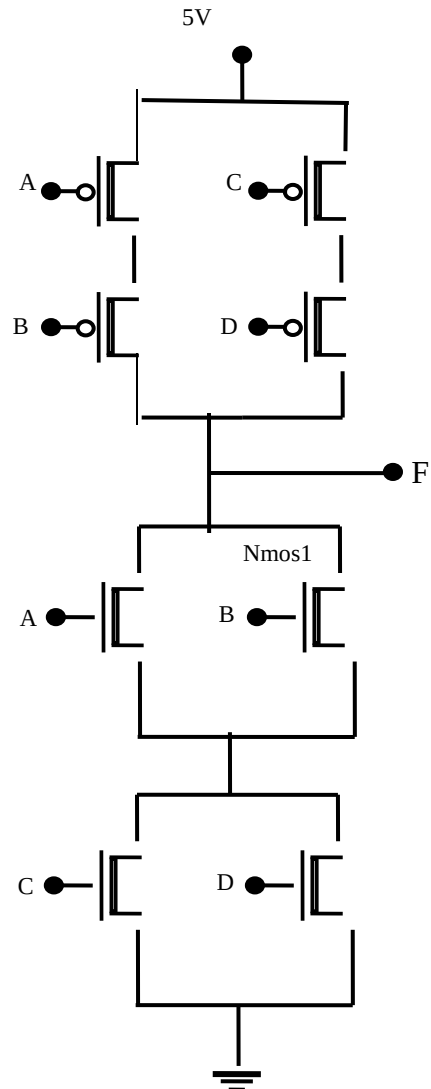
N must be an integer less than or equal to 63.3

Therefore take $N = 63$

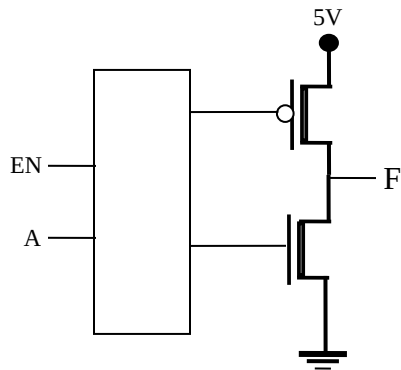
$$Fanout = \min\{86, 63\} = 63$$

6. (12 points) Draw the internal circuit of a CMOS circuit (using NMOS and PMOS transistors) which has the logic function $F = A' \cdot B' + C' \cdot D'$.

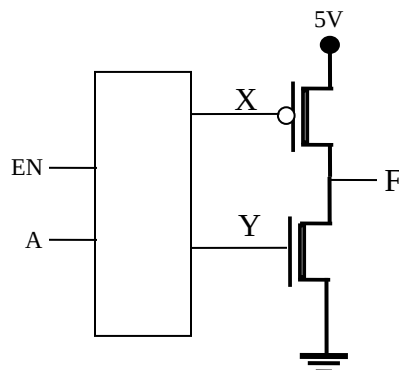
Solution: $F = A' \cdot B' + C' \cdot D' = [(A+B) \cdot (C+D)]'$



7. (16 points) The rectangular block in the below drawing represents a logic circuit (logic block). Design this logic circuit so that if EN is 0 then F is in Hi-Z state and if EN is 1 then $F = A'$. In other words, you are to design the inside of the logic block so that the whole circuit becomes a three-state inverter. Make your design using gates (not transistors). Draw the whole circuit.



Solution:



EN	A	X	Y
0	0	1	0
0	1	1	0
1	0	0	0
1	1	1	1

$$X = EN' + A$$

$$Y = EN \cdot A$$

